

# Bakary Gibba

AI Engineer | Applied AI Researcher

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[GitHub](#) • [LinkedIn](#) • [Google Scholar](#) • [ResearchGate](#) • [Portfolio](#)

LLMs • Agentic AI • Multi-Agent Systems • Backend Engineering

## Profile

Computer Science student and AI engineer with experience developing production AI applications, multi-agent systems, Retrieval-Augmented Generation (RAG), workflow automation, and intelligent backend services. Contributed to customer-facing AI products through backend engineering, LLM integration, and scalable software architecture using NestJS, Python, TypeScript, OpenAI, and Gemini. Active researcher in explainable AI and multi-agent resume-job matching, with a strong interest in building reliable AI systems that bridge research and real-world deployment.

## Research Areas

Multi-Agent Systems • Agentic AI • Retrieval-Augmented Generation (RAG) • Explainable AI • Large Language Models • Intelligent Automation • Human-Centered AI • AI for Healthcare

## Education

**Albukhary International University**

Kedah, Malaysia

Bachelor of Computer Science (Hons.)

Expected: Dec. 2026

CGPA: **3.50/4.00**

## Professional Experience

**AI Engineering Intern**

April 2026 – Present

*Apping Technology Sdn. Bhd., Kuala Lumpur, Malaysia*

- Developed production AI platforms including **Compass**, **AI HR Assistant**, **Closer AI**, and intelligent automation systems by building scalable backend services, REST APIs, and AI-powered workflows.
- Engineered LLM applications using OpenAI, Gemini, Retrieval-Augmented Generation (RAG), vector databases, prompt engineering, and configurable multi-provider architectures to support enterprise knowledge retrieval and content generation.
- Collaborated with cross-functional teams to deliver customer-facing AI features, automate business processes, improve system reliability, and deploy production-ready software using NestJS, TypeScript, Python, PostgreSQL, Docker, and Git.

## Research Experience

**Undergraduate Researcher**

2024 – Present

*Albukhary International University, Malaysia*

- Conduct research in Explainable AI, medical image analysis, multi-agent systems, and intelligent decision-support systems in collaboration with faculty members and international research partners.
- Design and evaluate deep learning, hybrid AI, and machine learning models using CNNs, DenseNet, U-Net, SVM, and Explainable AI techniques (Grad-CAM, SHAP) for healthcare and real-world AI applications.
- Contribute to peer-reviewed journal publications, conference papers, and open research initiatives as first author, co-first author, and collaborating researcher.

## Selected Publications & Preprints

- Md Serajun Nabi, Md Rashidul Islam, **Bakary Gibba**, et al. *Skin Lesion Classification with Explainable AI to Enhancing Dermatological Diagnosis*. Proceedings of the IEEE Multimedia University Engineering Conference (MECON), 2025. DOI: [10.1109/MECON67253.2025.11277079](#)
- Md Rashidul Islam, **Bakary Gibba**, Altagi Abdallah Bakheit Abdelgadir. *A Hybrid Deep Learning Framework with Explainable AI for Lung Cancer Classification with DenseNet169 and SVM*. Under review, IAES International Journal of Artificial Intelligence (IJ-AI), 2025. Preprint: [arXiv:2512.03359](#)
- Md Rashidul Islam\*, **Bakary Gibba\***, et al. *An Explainable AI-Driven Framework for Automated Brain Tumor*

*Segmentation Using an Attention-Enhanced U-Net*. arXiv Preprint, 2026. DOI: [10.48550/arXiv.2603.23344](https://doi.org/10.48550/arXiv.2603.23344)

4. Md Rashidul Islam\*, **Bakary Gibba\***, et al. *Machine Learning Approaches for Chronic Kidney Disease Prediction and Pattern Analysis*. Under review, Journal of Engineering and Technology and Applied Physics, 2026.
5. **Bakary Gibba**, et al. *Multi-Agent Machine Learning Framework for Explainable Resume–Job Matching Using LLM-Based Agents*. Under review, 2026.
6. Prof. Dr. Zurinahni Zainol, **Bakary Gibba**, Altagi Abdallah Bakheit Abdelgadir, et al. *Phone Usage, Study Habits, and Academic Performance: Predicting GPA with Machine Learning*. Presented at the 10th International Conference on Computing and Informatics (ICOCI), 2025.
7. Vahid Mato, **Bakary Gibba**, et al. *Trends and Determinants of Early Initiation and Exclusive Breastfeeding Rates from the Gambia Demographic Health Survey*. Under review, Scientific Reports (Springer Nature), 2025.

## Selected Engineering Portfolio

### Compass – AI Marketing Intelligence Platform Inprogress

AI-powered marketing platform for client intelligence, content generation, workflow automation, and production planning using multi-provider LLMs. *Technologies: NestJS, TypeScript, PostgreSQL, OpenAI, Gemini*

### AI HR Assistant GitHub

Enterprise Retrieval-Augmented Generation (RAG) assistant supporting intelligent document search, knowledge retrieval, and configurable AI providers. *Technologies: NestJS, Qdrant, PostgreSQL, OpenAI, Gemini*

### Closer AI GitHub

Production AI sales assistant integrating conversational AI, backend services, and workflow automation to improve customer engagement. *Technologies: TypeScript, NestJS, PostgreSQL, OpenAI*

### Rednote Automation GitHub

AI-driven content automation pipeline for social media publishing using browser automation and intelligent workflow orchestration. *Technologies: Python, n8n, Appium, LLM APIs*

## Technical Expertise

|                                  |                                                                                                                                                                              |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Languages                        | Python, TypeScript, JavaScript, SQL, R                                                                                                                                       |
| AI & Machine Learning Frameworks | Large Language Models (LLMs), Multi-Agent Systems, Retrieval-Augmented Generation (RAG), Prompt Engineering, Explainable AI (SHAP, Grad-CAM), Deep Learning, Computer Vision |
| Databases                        | NestJS, Node.js, Express.js, Prisma ORM, Hugging Face Transformers                                                                                                           |
| Tools & Platforms                | PostgreSQL, MongoDB, Qdrant, Vector Databases                                                                                                                                |
|                                  | Git, GitHub, Docker, Linux, Postman, n8n, Appium, OpenAI API, Gemini API                                                                                                     |

## Awards & Recognition

- Albukhary Foundation Scholarship (Fully Funded Undergraduate Scholar)
- Dean's List Award (4×), Albukhary International University
- Best Research Presenter Award, AIU Research Symposium
- International Conference Presenter (AI & Machine Learning)
- Maybank Global #MBassador, ASEAN Youth Leadership Programme

## Professional Certifications

- IBM AI Engineering Professional Certificate Link
- IBM Data Science Professional Certificate Link
- IBM Data Analyst Professional Certificate Link